

ARYAN BHAPKAR

Machine Learning Engineer | Robotics & Autonomous Systems

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Robotics and AI Engineer with an M.S. from Arizona State University and hands-on research experience at the Interactive Robotics Laboratory, with four peer-reviewed IEEE publications in computer vision, biometric authentication, and industrial robotics. Experienced in developing and deploying sim-to-real machine learning systems, including a 5-DOF robotic arm achieving 95% real-world accuracy with 30ms latency. Strong expertise in deep learning, reinforcement learning, and ROS2, with a focus on transforming research prototypes into reliable, production-ready systems.

EDUCATION

Arizona State University - Tempe, AZ

Aug 2024 - Dec 2025

Master of Science (M.S.) in Robotics and Autonomous Systems (Artificial Intelligence)

NMIMS Mukesh Patel School of Technology Management & Engineering - Mumbai, India

Aug 2020 - May 2024

B. Tech in Artificial Intelligence

TECHNICAL SKILLS AND CERTIFICATIONS

Machine Learning and AI: Deep Learning, Reinforcement Learning, CNNs, GANs, LLMs, Fine-Tuning, MLOps, NLP, Model Training, Data Preprocessing, Model Evaluation, Supervised & Unsupervised Learning, RAG

Robotics & Control: ROS2, MuJoCo, URDF, Robot Kinematics, Trajectory Planning, Kalman Filtering, Dynamic Movement Primitives (DMPs), Inverse Kinematics (IK)

Computer Vision: OpenCV, YOLOv9, Optical Flow, Feature Extraction, Object Detection, Image Preprocessing, Visual Tracking

Programming: Python, SQL, Arduino C, R, Bash

Frameworks & Tools: NumPy, Pandas, SciPy, Scikit-Learn, PyTorch, TensorFlow, AWS (EC2, Lambda, S3, DynamoDB), Matplotlib, Linux, MATLAB, ROS2 Gazebo, REST APIs, Git

WORK EXPERIENCE

Apex Systems, San Francisco, CA:

July 2026 - Present

Robotics Operations Technician - OpenAI Robotics Lab

- Supported **robotic system testing and data-collection** workflows by operating robots, executing predefined motion routines, and documenting experimental observations and system behavior.
- Assisted with **hands-on robot setup, equipment preparation, and laboratory operations**, gaining practical experience with physical robotic hardware and real-world robotics testing workflows.

Interactive Robotics Laboratory, Tempe, AZ:

Aug 2025 - July 2026

Research Aide | (Advisors: Prof. Heni Ben Amor & Mohamed Mistiri)

- Designed and deployed a high-fidelity **MuJoCo simulation of a UR5e manipulator** for adaptive tabletop manipulation, integrating reinforcement learning policies with Cartesian trajectory planning to enable robust contact-rich behavior
- Architected an **LLM-driven robot control stack using Google Gemini Flash 2.5** that converts free-form natural language commands into executable, real-time motion primitives closing the gap between human intent & embodied robot action

PROJECTS

5-DOF Robotic Arm Table Tennis Prototype

Spring 2025

- Engineered an end-to-end **sim-to-real pipeline in Gazebo / ROS2** combining YOLOv9 ball detection, optical flow, Kalman filtering, and a **TD3 reinforcement-learning policy** for closed-loop trajectory control.
- Achieved **95% real-world hit accuracy at 30ms decision-to-actuation latency**, demonstrating successful zero-shot transfer of the trained policy from simulation to physical hardware.

StyleGAN-Based Style Mapper

Fall 2023

- Built a high-resolution, **single-reference style-transfer system** leveraging StyleGan, GAN inversion, and JojoGAN mapping arbitrary photographic inputs to user defined artistic styles with photo-realistic fidelity

PEER - REVIEWED PUBLICATIONS (IEEE)

- Mitigating the Effect of Shadows in Vision-Based Industrial Robot** - IEEE ICETA, 2024 ([DOI](#))

Shadow-resilient detection pipeline (lighting optimization + Patmax) deployed on the KUKA KR4 AGILUS for industry.

- Color Sorting Pick Up and Drop Robotic Arm** - IEEE I2CT2 2024 ([DOI](#))

Sensor-driven color-sorting pick and place robotic arm for automated industrial sorting workflows.

- EarSiamNet: Leveraging Siamese Networks & SVM for Ear Biometric Authentication** -IEEE MoSICOM 2023 ([DOI](#))

Siamese-network embeddings with HOG features and SVM classification - 97.86% authentication accuracy.

- Open Set Masked Face Identification System** - IEEE ICCVMI Spring 2023 ([DOI](#))

Open-set identification system robust to face occlusion relevant to post-pandemic identity verification.